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Revisiting Momentum Crashes: Volatility Timing across Developed and Emerging Markets

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Abstract

This paper revisits momentum crash risk in developed and emerging equity markets using monthly Fama-French international factor data through April 2026. We distinguish conditional momentum-return regressions from direct crash-probability models, define the investment rules completely, and impose strictly ex-ante portfolio scaling. Lower-tail momentum events cluster after weak market states, but the evidence is more precise in developed markets and weakens under alternative definitions and after 2010. Volatility management materially improves full-sample downside outcomes: the developed-market Sharpe ratio rises from 0.533 to 0.865 and maximum drawdown improves from -43.61% to -10.74%; the emerging-market Sharpe rises from 0.980 to 1.289. These gains are not uniform over time, particularly in emerging markets after 2010. The results support state-aware momentum risk control while cautioning against interpreting conditional associations as stable forecasting laws.

Keywords: momentum crashes; volatility management; market states; factor investing; emerging markets; downside risk

JEL Classification: G11, G12, G15, G17

1. Introduction

Momentum is one of the most persistent empirical regularities in asset pricing, yet it is also one of the most difficult factor exposures to hold through adverse states. The basic return continuation documented by Jegadeesh and Titman (1993) has survived alternative samples, markets and portfolio constructions, and international evidence shows that the phenomenon is not confined to US equities (Rouwenhorst 1998; Jegadeesh and Titman 2001; Asness et al. 2013). Momentum also remains difficult to absorb into standard factor models, which is why Carhart (1997) treated it as a distinct source of return and why later work continues to examine both its economic origin and its implementation risk (Fama and French 1996; Ehsani and Linnainmaa 2022; Arnott et al. 2023; Feng et al. 2026).

The difficulty is that a positive unconditional premium can coexist with unusually severe left-tail losses. Grundy and Martin (2001) show that momentum portfolios carry strongly time-varying exposures, while Cooper et al. (2004) establish that momentum profits depend on prior market states. Daniel and Moskowitz (2016) provide the central crash mechanism used in this paper: following prolonged market weakness, past losers can acquire high conditional market exposure and then rebound sharply when aggregate markets recover. A long-winner/short-loser portfolio is vulnerable precisely because its short leg can rally violently. The resulting losses are episodic rather than continuously high, which makes ordinary volatility or mean-return comparisons insufficient measures of implementation risk. Long historical evidence and more recent work on crash isolation reinforce the importance of treating momentum as a state-dependent payoff rather than as a constant premium (Geczy and Samonov 2016; Dierkes and Krupski 2022).

Several economic interpretations are consistent with state dependence. Behavioural models connect momentum to delayed reaction, overconfidence and gradual information diffusion (Barberis et al. 1998; Daniel et al. 1998; Hong and Stein 1999). Risk-based interpretations instead relate momentum to time-varying expected returns and business-cycle exposure (Chordia and Shivakumar 2002; Griffin et al. 2003). These mechanisms need not be mutually exclusive for the applied question studied here. Whether the underlying source is behavioural, risk-based or a combination, investors face a practical problem: the distribution of momentum returns is not stable across market states, and the states associated with large reversals may be partly observable before the return is realised.

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A related literature asks whether volatility scaling can convert momentum into a more investable exposure. Barroso and Santa-Clara (2015) show that momentum risk varies substantially over time and that inverse-variance management can reduce crash exposure. Moreira and Muir (2017) generalise volatility management across factor portfolios. However, the literature also warns against reading in-sample improvements as a universal timing principle (Feng 2026). Asness et al. (2017) emphasise the difficulty of factor timing, Cederburg et al. (2020) document substantial variation in volatility-management performance across specifications, and DeMiguel et al. (2024) show that the evaluation becomes more demanding in a multifactor setting. Kim et al. (2016) further demonstrate that volatility scaling itself can account for a large part of apparent time-series momentum performance. These findings motivate a deliberately transparent design in which the risk-management rule is simple, lagged and fully reproducible.

The international dimension is useful for two reasons. First, most of the classic crash evidence is developed from US or broad developed-market settings, while global asset allocators frequently use regional or international factor portfolios. Second, implementation conditions differ materially between developed and emerging markets. Trading costs, short-sale constraints, liquidity, market impact and capacity can erode theoretical momentum profits, and those frictions are often stronger in less liquid markets (Lesmond et al. 2004; Korajczyk and Sadka 2004; Israel and Moskowitz 2013). A comparison of developed and emerging factor returns therefore provides an informative test of whether the same observable state variables carry similar information across market groups, while avoiding the stronger claim that aggregate factor returns are directly tradable net of all frictions.

The paper addresses three questions. First, how are subsequent momentum returns related to lagged bear-market conditions and market volatility, and does a direct binary model indicate that those variables also contain information about the probability of a lower-tail momentum event? Second, how precise are the differences between developed and emerging markets? Third, do transparent volatility-managed and dynamic market-state rules improve downside outcomes when all portfolio inputs and scaling constants are constructed without future information?

The paper does not propose a new momentum anomaly or a new explanation for the 2009 reversal. Its contribution is instead to test whether the established state-dependent crash mechanism remains visible in public international data, whether direct lower-tail models support the evidence from

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continuous-return regressions, and whether transparent ex-ante risk controls remain economically useful across developed and emerging markets.

Predictability is interpreted cautiously. Conditional return variation, higher estimated crash probability and operational forecasting are distinct concepts. Lagged state variables may explain next-month WML or identify higher-risk states without reliably forecasting rare events out of sample, and inference can depend on whether crashes are defined using 5%, 10% or rolling thresholds. The paper therefore distinguishes these levels of evidence and avoids treating in-sample associations as stable forecasting laws, particularly because empirical asset-pricing results can be sensitive to specification and sample choices (Harvey et al. 2016; Hou et al. 2020; Jensen et al. 2023; Menkveld et al. 2024).

The results point to a market- and specification-dependent pattern. In developed markets, the bear-state × volatility interaction is significantly negative, and the 5% crash logit indicates that lagged market conditions contain information about subsequent tail events. Emerging-market coefficients are much less precise. Descriptively, 5% crashes occur in 16.67% of developed bear-state observations versus 2.98% of non-bear observations, compared with 8.63% and 2.56% in emerging markets. However, alternative crash definitions and subperiod tests show that the relationship is not a stable forecasting law, with weaker post-2010 evidence.

Volatility management improves full-sample downside outcomes in both market groups. The developed-market Sharpe ratio rises from 0.533 to 0.865 and maximum drawdown improves from -43.61% to -10.74%; in emerging markets, the Sharpe ratio rises from 0.980 to 1.289 and maximum drawdown falls from -37.26% to -15.45%. The gains are stronger before 2008 and weaken after 2010, particularly in emerging markets, where the post-2010 managed Sharpe ratio of 1.290 remains below the static value of 1.356.

The contribution is therefore not a claim of a new universal momentum-crash law. Instead, the paper provides updated international evidence, directly separates conditional returns from crash probabilities, and shows how state information and volatility scaling can be implemented with transparent, reproducible rules. This positioning is consistent with the broader emphasis on specification discipline and replication in empirical finance (Harvey et al. 2016; Hou et al. 2020; Jensen et al. 2023; Menkveld et al. 2024). The remainder of the paper describes the data, formalises

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the empirical and portfolio methods, reports the results and robustness tests, and discusses the implications for global factor allocation.

2. Data

The analysis uses monthly international factor returns from the Kenneth R. French Data Library files supplied with the **replication package**[1]. The full replication package, including a one-click Python Jupyter Notebook for reproducing the empirical analyses, is available at the GitHub Link. Four raw files are combined: Developed Markets five factors, Developed Markets momentum, Emerging Markets five factors and Emerging Markets momentum. The five-factor files contain the market excess return (Mkt-RF), size (SMB), value (HML), profitability (RMW), investment (CMA) and the one-month risk-free rate. The momentum files contain winner-minus-loser (WML) returns. The factor choice allows the momentum premium to be studied separately from the standard market, size, value, profitability and investment dimensions associated with Fama and French (1993, 2015).

The raw files require a small but important cleaning step. Textual metadata above the header are discarded, -99.99 is treated as missing, and only six-digit YYYYMM values are retained as monthly observations. This last rule matters because the developed five-factor CSV appends annual January-December summary rows after the monthly section. Those annual rows are not observations in the monthly regression sample and are therefore removed automatically. All percentage-point returns are converted to decimals after cleaning.

Momentum availability determines the static WML samples. Developed-market momentum begins in November 1990 and ends in April 2026, giving 426 observations. Emerging-market momentum begins in January 1990 and ends in April 2026, giving 436 observations. The ten-observation difference in the static strategies is therefore exactly the ten-month difference in momentum start dates; it is not caused by strategy construction or arbitrary deletion. The emerging five-factor regression uses 406 observations because early RMW and CMA values are missing in the five-factor source, with CMA determining the common complete-case start date.

[1] https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/AE_Momentum_Crashes_across_Developed_and_Emerging_Markets

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The use of market-group factors is intentional. Aggregate developed and emerging factors avoid country-selection discretion and provide consistent market-group comparisons, at the cost of obscuring country- and security-level drivers of individual crashes.

Data alignment is performed by calendar month after the four source files are cleaned separately. No interpolation is used for factor returns. A regression observation is retained only when the dependent variable and all regressors required by that specification are available. This rule explains why the five-factor sample can be shorter than the static WML sample without affecting the latter. It also avoids silently backfilling the early emerging RMW or CMA history.

WML is the return difference between the past-winner and past-loser portfolios. Table 1 reports annualised mean return and volatility, Sharpe and Sortino ratios, skewness and maximum drawdown. The subperiod split is deliberately coarse: Pre-2008 ends in December 2007; Post-2010 begins in January 2010; 2008-2009 is treated as crisis/transition and excluded from the subperiod comparison. This separation avoids using the global financial crisis itself to define both sides of a pre/post comparison.

Table 1. Momentum performance: full sample and subperiods

Panel A. Full sample

Market	N	Mean ann.	Vol ann.	Sharpe	Sortino	Skewness	Max DD
Developed	426	6.85%	12.85%	0.533	0.601	-0.932	-43.61%
Emerging	436	10.24%	10.45%	0.980	1.106	-0.991	-37.26%

Panel B. Pre-2008 and post-2010

Market	Period	N	Mean ann.	Vol ann.	Sharpe	Skewness	Max DD
Developed	Pre-2008	206	9.87%	13.69%	0.721	-0.302	-43.05%
Developed	Post-2010	196	6.45%	9.57%	0.674	-0.367	-20.10%
Emerging	Pre-2008	216	10.94%	10.35%	1.056	-0.679	-21.61%
Emerging	Post-2010	196	12.48%	9.20%	1.356	-0.224	-14.52%

Notes: Crisis/transition months (2008-2009) are excluded from Panel B.

The descriptive evidence establishes the central economic trade-off. Emerging-market WML has the higher full-sample annualised mean, 10.24% versus 6.85%, and the higher Sharpe ratio, 0.980 versus 0.533. Both series nevertheless display negative skewness and severe drawdowns. Developed WML loses 43.61% from peak to trough, while emerging WML loses 37.26%. After 2010, downside risk is milder in both groups, especially in developed markets, but negative skewness remains. These factor-level returns should not be read as net implementable profits. Transaction costs and shorting frictions are omitted and can be especially material in emerging markets (Lesmond et al. 2004; Korajczyk and Sadka 2004; Israel and Moskowitz 2013).

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3. Methodology

3.1 Market states and crash definitions

Let $WML_{m,t}$ denote the monthly momentum factor return for market group $m \in \{D, E\}$, where D is developed and E is emerging. The baseline lower-tail crash indicator is defined using the market-specific empirical q-th percentile $Q_m(q)$:

$$C_{m,t}(q) = \mathbb{1}\big[WML_{m,t} \le Q_m(q)\big], \qquad q = 0.05. \qquad (1)$$

The 5% definition is retained as the baseline because it focuses on economically unusual reversals rather than ordinary negative momentum months. At the same time, the event is rare by construction, so coefficient precision can be sensitive to a small number of observations. The 10% threshold provides a broader downside definition with more events, while the rolling 60-month threshold asks a different question: whether a month is unusually poor relative to the distribution an investor could have observed immediately beforehand. Agreement across these definitions would be strong evidence; disagreement is itself informative and is reported rather than hidden. None of the threshold choices is selected by maximising a strategy Sharpe ratio or by searching for the strongest t-statistic.

The 10% tail is used as an alternative definition. Equation (1) identifies realised lower-tail events directly, but the threshold $Q_m(q)$ is estimated from the unconditional sample. Consequently, the baseline crash label is suitable for a retrospective probability model but should not be interpreted as a fully real-time operational threshold. To remove that concern in robustness tests, we also use a rolling 60-month quantile based only on returns available through t-1:

$$C^{roll}_{m,t}(0.05) = \mathbb{1}\big[WML_{m,t} \le Q^{60}_{m,t-1}(0.05)\big]. \qquad (2)$$

Market stress is measured from the total market return $R^M_{m,t}$, constructed as Mkt-RF plus RF. The lagged 24-month cumulative market return is

$$R^{M,24}_{m,t-1} = \prod_{j=1}^{24}\big(1 + R^M_{m,t-j}\big) - 1. \qquad (3)$$

The bear-state indicator is

$$Bear_{m,t-1} = \mathbb{1}\big[R^{M,24}_{m,t-1} < 0\big]. \qquad (4)$$

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This definition follows the market-state logic in Cooper et al. (2004) and Daniel and Moskowitz (2016). A negative 24-month market history is not itself a claim that a crash will occur; it identifies a state in which the composition and beta exposure of winner and loser portfolios may be unusually important.

Lagged market volatility is the sample standard deviation of monthly market excess returns over the previous h months:

$$Vol^{M}_{m,t-1}(h) = sd\left(MktRF_{m,t-1}, \ldots, MktRF_{m,t-h}\right). \qquad (5)$$

The baseline uses $h = 12$, with 6- and 24-month windows as robustness checks. Momentum variance for portfolio scaling is estimated analogously from the previous 12 WML observations.

3.2 Conditional momentum-return regression

The first specification estimates how the conditional mean of WML varies with prior market states. It estimates how the conditional mean of WML varies with prior market state:

$$WML_{m,t} = \alpha_m + \beta_{1m} Bear_{m,t-1} + \beta_{2m} Vol^{M}_{m,t-1} + \beta_{3m}\left(Bear_{m,t-1} \times Vol^{M}_{m,t-1}\right) + \varepsilon_{m,t}. \qquad (6)$$

A negative β_{3m} indicates that high volatility is associated with weaker subsequent momentum returns when the market has also experienced a negative 24-month state. Equation (6) is a conditional-return model. It does not by itself estimate the probability of a crash, and the empirical discussion therefore avoids calling Table 3 Panel A a crash-predictability regression. OLS inference uses Newey-West heteroskedasticity- and autocorrelation-consistent covariance estimates with six lags (Newey and West 1987).

The coefficients in Equation (6) are interpreted as descriptive conditional associations rather than structural causal effects. The bear indicator and volatility are functions of realised market history, not randomly assigned treatments, and the specification does not attempt to isolate a causal channel from all other macroeconomic or liquidity conditions. The interaction is nevertheless economically useful because it tests a precise implication of the crash literature with information dated before the WML return. Six HAC lags are used consistently across the monthly return regressions; changing the volatility measurement window is handled separately in robustness tests so that inference and state construction are not conflated.

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3.3 Direct crash-probability model

To test the crash indicator directly, the paper estimates a logit model:

$$Pr\left(C_{m,t}(0.05) = 1 \mid \mathcal{F}_{t-1}\right) = \Lambda\left[\gamma_{0m} + \gamma_{1m} Bear_{m,t-1} + \gamma_{2m} Vol^{M}_{m,t-1} + \gamma_{3m} Bear_{m,t-1} Vol^{M}_{m,t-1}\right], \quad (7)$$

where $\Lambda(z) = 1/(1 + e^{-z})$ and $\mathcal{F}_{t-1}$ denotes information available before month t. Equation (7) directly addresses whether lagged state variables contain information about a subsequent lower-tail WML event. HAC covariance estimates with six lags are used for the reported test statistics. Because logit coefficients are in log-odds units, the discussion focuses on signs, statistical precision, fitted probabilities and state-specific crash frequencies rather than treating the raw coefficient magnitude as a probability change.

This distinction is particularly important in the presence of an interaction. For a bear state, the effect of volatility on log odds is $\gamma_{2m} + \gamma_{3m}$ rather than γ_{2m} alone, while the effect of the bear indicator depends on the prevailing volatility level. Consequently, a positive main bear coefficient and a negative interaction can still imply a higher fitted crash probability over the empirically relevant volatility range. The paper therefore supplements coefficient tables with fitted-probability curves and simple crash frequencies by state. These quantities are easier to interpret economically and reduce the risk of reading one logit coefficient in isolation.

The 10% unconditional tail and the rolling 60-month 5% tail in Equation (2) are re-estimated as robustness checks. The rolling version is particularly important for interpretation: both the predictors and the crash threshold are then based entirely on data available by t-1. If conclusions weaken under that stricter definition, the appropriate inference is that crash-risk information is specification-sensitive rather than that a strong forecasting law has been established.

3.4 Rebound mechanism

Daniel and Moskowitz (2016) emphasise that momentum crashes occur not simply during market declines but during strong rebounds after weak states. To examine that mechanism, define $Up_{m,t} = 1$ when the contemporaneous market excess return is positive. We estimate

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$$WML_{m,t} = a_m + b_{1m}MktRF_{m,t} + b_{2m}Bear_{m,t-1} + b_{3m}\left(Bear_{m,t-1}MktRF_{m,t}\right) + b_{4m}\left(Bear_{m,t-1}Up_{m,t}MktRF_{m,t}\right) + u_{m,t}. \quad (8)$$

The final interaction asks whether positive market rebounds after bear states are associated with especially low momentum returns. Because Equation (8) includes contemporaneous market returns, it is a mechanism regression rather than an ex-ante forecasting model. This distinction is important for interpreting the results.

3.5 Five-factor alpha and cross-market comparison

To verify that WML remains economically meaningful after standard factor controls, the paper estimates

$$WML_{m,t} = \alpha_m + \beta_M MktRF_{m,t} + \beta_S SMB_{m,t} + \beta_H HML_{m,t} + \beta_R RMW_{m,t} + \beta_C CMA_{m,t} + \epsilon_{m,t}. \quad (9)$$

The factor set follows Fama and French (2015), while the interpretation of momentum as a separate return source is connected to the earlier anomaly evidence in Fama and French (1996) and Carhart (1997).

Cross-market differences are examined by stacking developed and emerging observations and interacting the state variables with an emerging-market indicator EM_m:

$$WML_{m,t} = \delta_0 + \delta_1 EM_m + \delta_2 Bear_{m,t-1} + \delta_3 Vol^M_{m,t-1} + \delta_4 Bear_{m,t-1}Vol^M_{m,t-1} + \delta_5 EM_m Bear_{m,t-1} + \delta_6 EM_m Vol^M_{m,t-1} + \delta_7 EM_m Bear_{m,t-1}Vol^M_{m,t-1} + e_{m,t}. \quad (10)$$

Insignificant emerging-market interactions are not interpreted as proof of equality. Instead, it reports the individual interaction estimates and a joint Wald test of $H_0: \delta_5 = \delta_6 = \delta_7 = 0$. Failure to reject this null means only that the sample does not provide sufficient evidence to distinguish the coefficients at conventional levels.

3.6 Volatility-managed strategy

Let $\hat{\sigma}^2_{m,t-1}$ be the variance of WML estimated from the previous 12 monthly observations. The raw inverse-variance signal is $1/\hat{\sigma}^2_{m,t-1}$. The scaling constant is determined recursively rather than from the full sample. For each month t, define the expanding normalisation

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$$c_{m,t}^{vol} = \left[\frac{1}{N_t}\sum_{s\le t}\frac{1}{\hat{\sigma}_{m,s-1}^2}\right]^{-1}, \qquad (11)$$

where the sum includes only months with an available lagged variance. Since $\hat{\sigma}_{m,t-1}^2$ is known before $WML_{m,t}$ is realised, Equation (11) uses no future return information. The volatility-managed weight is

$$w_{m,t}^{vol} = \min\left(3, \frac{c_{m,t}^{vol}}{\hat{\sigma}_{m,t-1}^2}\right), \qquad r_{m,t}^{vol} = w_{m,t}^{vol} WML_{m,t}. \qquad (12)$$

The leverage cap of three prevents extreme exposures when estimated variance is unusually low. The expanding normalisation keeps the rule close to unit average raw exposure while retaining a genuinely ex-ante construction. It also makes the choice of the scaling constant transparent and directly reproducible.

The timing convention is important. The month-t weight is computed after month t-1 has closed and before the month-t WML return is applied. The expanding normaliser includes the current month's inverse-variance signal because that signal itself is a function only of returns through t-1; it never uses $WML_{m,t}$. The cap of three is fixed in advance for both market groups and for all subperiods. We do not optimise the cap separately by market, and we do not rescale the realised return series afterwards to match the volatility of static WML. These choices make the strategy comparison less favourable to ex-post performance engineering and more transparent for replication.

3.7 Dynamic market-state strategy

The dynamic rule uses the state model to form an expanding-window expected WML return. Before each month t, Equation (6) is re-estimated using all valid observations strictly before t. At least 60 complete historical regression observations are required. The resulting one-step-ahead fitted mean is

$$\hat{\mu}_{m,t} = \hat{\alpha}_{m,t-1} + \hat{\beta}_{1,m,t-1} Bear_{m,t-1} + \hat{\beta}_{2,m,t-1} Vol_{m,t-1}^{M} + \hat{\beta}_{3,m,t-1} Bear_{m,t-1} Vol_{m,t-1}^{M}. \qquad (13)$$

The raw dynamic exposure is mean divided by lagged variance. Its scale is also determined recursively:

$$c_{m,t}^{dyn} = \left[\frac{1}{K_t}\sum_{s\le t}\left|\frac{\hat{\mu}_{m,s}}{\hat{\sigma}_{m,s-1}^2}\right|\right]^{-1}. \qquad (14)$$

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The final position and return are

$$w_{m,t}^{dyn} = clip\left(c_{m,t}^{dyn}\frac{\hat{\mu}_{m,t}}{\hat{\sigma}_{m,t-1}^{2}}, -3,3\right), \qquad r_{m,t}^{dyn} = w_{m,t}^{dyn} WML_{m,t}. \qquad (15)$$

Every input in Equations (13)-(15) is observable at the portfolio-formation date. The dynamic sample begins only after enough history exists. The 24-month bear-state construction consumes the first 24 observations, and the expanding return regression then requires 60 complete historical state observations. Hence the first dynamic return appears after an 84-month burn-in. This explains exactly why developed N falls from 426 to 342 and emerging N from 436 to 352. The volatility-managed strategy requires only the 12-month variance window, giving 414 and 424 observations respectively.

The portfolio tests report annualised return, volatility, Sharpe and Sortino ratios, skewness, monthly VaR and CVaR, and maximum drawdown. They are intended as transparent factor-level risk-management experiments, not as net-of-cost trading backtests. No parameter is chosen by searching over the realised performance table.

4. Empirical results

4.1 Anatomy of large momentum losses

Figures 1 and 2 show that both momentum series compound positively over the long sample but with sharp interruptions. The developed series experiences its deepest drawdown around the global financial crisis. Emerging momentum also suffers substantial crisis-era and later reversals despite its higher long-run average return.

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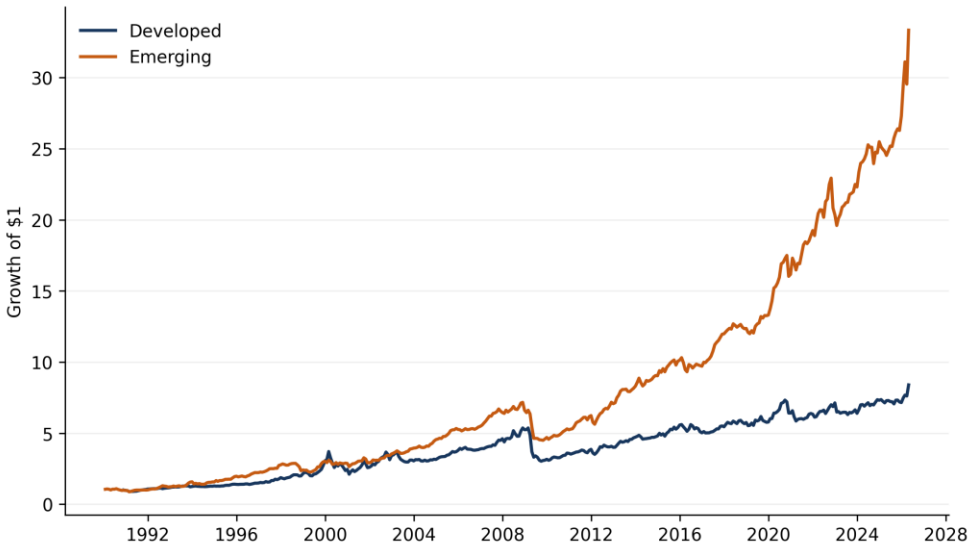


Figure 1. Cumulative wealth of static developed and emerging momentum portfolios.

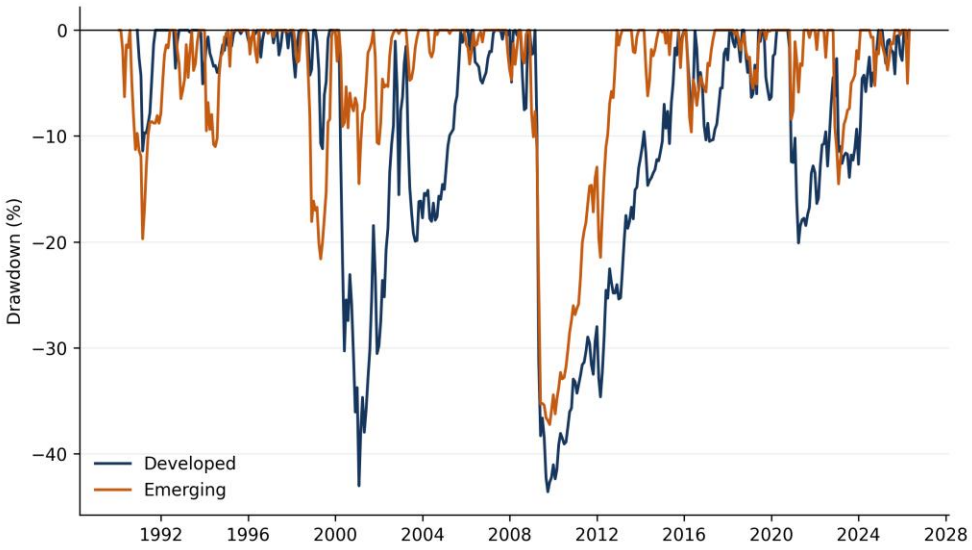


Figure 2. Drawdowns of static developed and emerging momentum portfolios.

Table 2 lists the eight worst WML months in each market group together with contemporaneous market returns and lagged state variables. The clearest common episode is 2009. Developed WML falls 24.26% in April 2009 while the market excess return rises 11.41% and the prior 24-month market return is -43.96%. May 2009 is also among the worst developed observations. Emerging WML loses 14.39% in April and 14.84% in May while its market rebounds by roughly 17%-18%. These observations fit the mechanism in Daniel and Moskowitz (2016): crash losses can arrive during recovery rather than during the preceding panic itself.

Table 2. Worst monthly WML returns and market states

Panel A. Developed

Date	WML	Mkt-RF	24m MKT lag	Bear	12m vol lag
2009-04	-24.26%	11.41%	-43.96%	1	8.00%
2001-01	-14.01%	1.87%	12.99%	0	4.37%
2000-03	-12.23%	4.66%	39.23%	0	3.65%
2000-05	-11.02%	-3.28%	31.52%	0	4.21%
2020-11	-10.92%	13.34%	16.86%	0	6.80%
2000-04	-10.76%	-5.77%	40.49%	0	3.72%
2002-11	-10.65%	5.14%	-34.48%	1	5.24%
2009-05	-10.40%	9.95%	-40.04%	1	8.85%

Panel B. Emerging

Date	WML	Mkt-RF	24m MKT lag	Bear	12m vol lag
2009-05	-14.84%	17.98%	-29.58%	1	12.54%
2009-04	-14.39%	16.98%	-36.78%	1	11.30%
1998-11	-10.82%	6.12%	-34.35%	1	9.72%
1994-01	-9.51%	-1.35%	96.87%	0	4.98%
2022-11	-9.15%	12.55%	-10.78%	1	3.64%
1991-02	-8.84%	12.31%			6.74%
2020-11	-8.42%	10.13%	19.43%	0	7.54%
2001-11	-8.25%	9.66%	-32.09%	1	7.24%

The table also prevents an overly narrow crisis interpretation. Developed reversals appear around the technology-market reversal, the 2020 reopening/rebound and later episodes. Emerging-market losses include late-1990s stress, the global financial crisis and more recent rebound months. The distribution is therefore state-dependent but not reducible to a single historical event.

4.2 Conditional returns versus direct crash probability

Table 3 distinguishes conditional momentum returns from direct crash probability. Panel A reports Equation (6) and is labelled as a conditional WML-return regression. In developed markets, the bear × volatility coefficient is -0.9672 with a test statistic of -2.02. The standalone volatility coefficient is not significant. This pattern implies that high market volatility is most damaging for the conditional mean of momentum when it occurs after a negative 24-month market state. In emerging markets the same interaction is negative (-0.2594) but statistically imprecise. The result therefore supports a stronger state interaction in developed markets without claiming that the emerging relationship is absent.

Panel B uses the actual 5% crash indicator as the dependent variable. For developed markets, the bear-state coefficient is positive and significant, the volatility coefficient is positive and highly significant, and the interaction is negative at the 10% level. Because the variables enter nonlinearly through the logit link, these coefficients should be read jointly rather than as separate probability changes. The descriptive frequencies are easier to communicate: among developed observations for which the bear indicator is available, 16.67% of bear-state months fall into the unconditional 5%

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WML tail, versus 2.98% of non-bear months. In emerging markets the corresponding rates are 8.63% and 2.56%, but the individual logit coefficients are not precisely estimated.

Table 3. Conditional momentum returns and direct crash probability

Panel A. Conditional WML-return regression

Variable	Developed	Emerging
const	0.0119***	0.0157***
	(2.78)	(3.49)
Bear_lag	0.0475*	0.0096
	(1.84)	(0.82)
Mkt_vol_12_lag	-0.1214	-0.0781
	(-0.94)	(-0.85)
Bear_x_MktVol	-0.9672**	-0.2594
	(-2.02)	(-1.26)
Observations	402	412
Adjusted R2	0.037	0.040

Panel B. Logit model for unconditional 5% WML-tail event

Variable	Developed	Emerging
const	-6.5513***	-4.1390***
	(-5.48)	(-4.04)
Bear_lag	3.0740**	-0.1730
	(1.99)	(-0.12)
Mkt_vol_12_lag	72.0048***	10.4351
	(4.05)	(0.54)
Bear_x_MktVol	-41.0435*	16.2607
	(-1.76)	(0.72)
Observations	402	412

*Notes: HAC test statistics in parentheses. ***, ** and * denote 1%, 5% and 10% significance.*

Figure 3 plots fitted 5% crash probabilities across the observed volatility range. The figure is not an out-of-sample performance claim. It shows how the fitted logit combines bear state and volatility. The developed relationship is visibly more structured, while the emerging curves are less sharply separated. This visual result matches the table: lagged state information is economically relevant, but its statistical form differs across market groups.

The magnitudes help clarify the nonlinearity. At the developed-market median lagged volatility of about 3.8% per month, the fitted 5% tail probability is approximately 2.1% outside a bear state and 9.1% in a bear state. At the 90th percentile of lagged volatility, roughly 5.8%, the corresponding fitted probabilities are about 8.8% and 15.9%. For emerging markets the separation is smaller and statistically less precise: at median lagged volatility the fitted probabilities are approximately 2.6% and 4.7%, rising to about 3.5% and 9.7% at the 90th percentile. These fitted values are illustrative in-sample probabilities, not calibrated forecasts, but they show why the interaction should be evaluated jointly rather than coefficient by coefficient.

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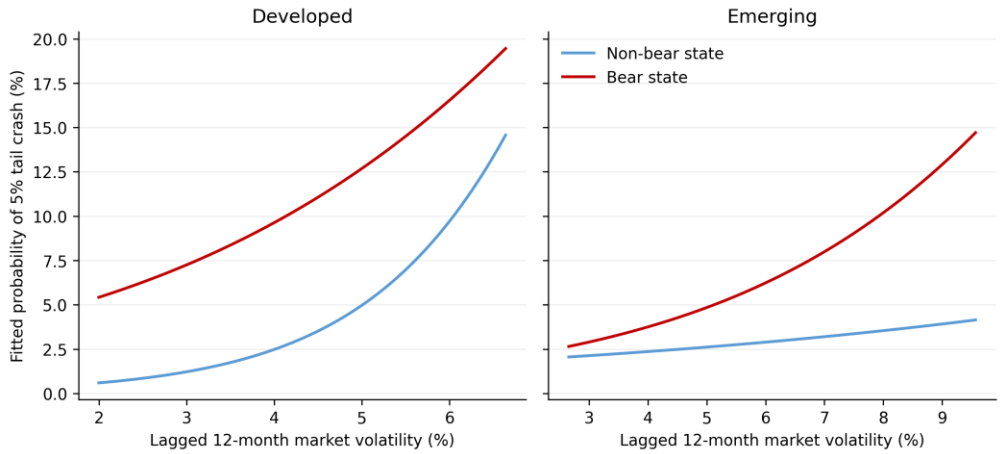


Figure 3. Fitted probability of a 5% momentum tail event across lagged market-volatility states.

The evidence therefore has two complementary layers. Equation (6) establishes an association between lagged states and the conditional mean of WML. Equation (7) directly relates those states to the probability of a tail event. Neither result, alone or together, justifies describing crashes as perfectly forecastable. The binary model uses lagged information, but the baseline tail threshold is defined using the unconditional sample distribution. The rolling-threshold robustness test below is therefore important for judging how much of the evidence survives a stricter real-time definition.

4.3 Rebound dynamics and market-group differences

Table 4 Panel A examines the rebound mechanism in Equation (8). Developed WML has a negative bear × market-return coefficient of -0.3035, marginally significant at 10%. The triple interaction with positive market months is negative but not significant. Emerging markets display a different pattern: the bear × up × market interaction is -0.4605 with a test statistic of -2.82. Thus the emerging evidence is strongest in the contemporaneous rebound mechanism rather than the lagged bear-volatility conditional-return regression. This distinction helps reconcile why dramatic emerging crash months are visible in Table 2 even though the ex-ante aggregate state regression is imprecise.

Panel B reports the pooled cross-market model. The baseline developed bear × volatility effect remains negative. None of the emerging-specific interaction coefficients is individually significant, and a joint Wald test of the three emerging-specific slopes produces a chi-square statistic of 2.414, for which the null cannot be rejected at conventional levels. The correct interpretation is limited: the

sample does not provide sufficient statistical evidence that the three slope adjustments differ jointly. It does **not** establish that developed and emerging market structures are identical. This wording matters because failure to reject equality is not evidence of exact equality.

Table 4. Rebound dynamics and cross-market comparison

Panel A. Conditional rebound regression

Variable	Developed	Emerging
const	0.0076***	0.0119***
	(4.31)	(8.11)
Mkt-RF	-0.0080	0.0421
	(-0.11)	(0.85)
Bear_lag	0.0074	0.0056
	(0.91)	(1.56)
Bear_x_Mkt	-0.3035*	-0.0854
	(-1.90)	(-0.94)
Bear_x_Up_x_Mkt	-0.5433	-0.4605***
	(-1.30)	(-2.82)
Observations	402	412
Adjusted R2	0.155	0.178

Panel B. Pooled developed-emerging regression

Variable	Pooled
const	0.0119***
	(2.78)
Emerging_dummy	0.0038
	(0.61)
Bear_lag	0.0475*
	(1.84)
Mkt_vol_12_lag	-0.1214
	(-0.94)
Bear_x_MktVol	-0.9672**
	(-2.02)
Emerging_x_Bear	-0.0379
	(-1.34)
Emerging_x_Vol	0.0433
	(0.27)
Emerging_x_Bear_x_Vol	0.7078
	(1.36)
Observations	814
Adjusted R2	0.039

*Notes: Joint Wald test of emerging-specific slopes: $\chi^2 = 2.414$ (not significant at 10%). HAC test statistics in parentheses. ***, ** and * denote 1%, 5% and 10% significance.*

4.4 Five-factor alpha

Table 5 confirms that the underlying WML exposure remains economically material after standard controls. Developed momentum earns a monthly intercept of 0.7048%, equivalent to an annualised alpha of approximately 8.46%, with a test statistic of 3.69. Emerging momentum earns a monthly intercept of 0.8242%, or about 9.89% annualised, with a test statistic of 4.41. The developed series has significant negative market and value loadings; the emerging series has a significant positive profitability loading. These loadings are not the main object of the paper, but they show that the

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momentum return studied here is not simply a relabelled combination of the five Fama-French factors.

Table 5. Fama-French five-factor alpha regressions

Variable	Developed	Emerging
const	0.0070***	0.0082***
	(3.69)	(4.41)
Mkt-RF	-0.1644***	-0.0286
	(-3.02)	(-0.55)
SMB	0.2693	0.1953
	(1.60)	(1.57)
HML	-0.6649***	-0.1041
	(-2.94)	(-0.96)
RMW	0.2124	0.3751***
	(0.84)	(2.69)
CMA	0.4053	0.2028
	(1.29)	(1.34)
Observations	426	406
Adjusted R2	0.168	0.076

*Notes: HAC test statistics in parentheses. ***, ** and * denote 1%, 5% and 10% significance.*

The five-factor results also motivate risk management rather than replacement. If WML were economically negligible after standard controls, reducing its crash exposure would have limited portfolio relevance. Instead, the alpha estimates are sizeable, consistent with the long literature treating momentum as a distinct anomaly or factor exposure (Fama and French 1996; Carhart 1997; Asness et al. 2013).

The alpha regressions are not used to claim that the WML return is a free-standing arbitrage profit. The intercepts describe average factor performance after linear exposure to the five controls in the same market group. They do not account for implementation costs, omitted nonlinear exposures or the possibility that a conditional model would attribute part of the average return to changing state risk. Their role is therefore deliberately limited: they confirm that the underlying momentum series remains economically nontrivial while the paper concentrates on the distribution and management of that return source.

4.5 Strategy performance with strictly ex-ante scaling

Table 6 Panel A reports the strategy results. The normalisation changes are important because the new scale at month t depends only on quantities observable by t. Developed static WML earns 6.85% annually with 12.85% volatility, a Sharpe ratio of 0.533 and a -43.61% maximum drawdown. Volatility management lowers the annualised mean slightly to 6.52%, but volatility falls to 7.54%. The Sharpe ratio therefore rises to 0.865, 5% monthly VaR improves from -5.01% to -2.28%, CVaR

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improves from -9.33% to -4.15%, and maximum drawdown falls dramatically to -10.74%. The result is best described as risk control rather than return enhancement.

The developed dynamic strategy earns 10.03% with a Sharpe ratio of 0.821. Its maximum drawdown, -26.18%, is much better than static WML but materially worse than the volatility-managed rule. This result is useful because it prevents the paper from presenting the more complex state rule as automatically superior. Once its estimation and scaling are made fully ex ante, the simpler volatility overlay is particularly effective at controlling developed-market tail risk.

Emerging-market results are stronger in the full sample. Volatility management raises annualised return from 10.24% to 15.65% and the Sharpe ratio from 0.980 to 1.289, while maximum drawdown falls from -37.26% to -15.45%. The dynamic strategy earns 11.31%, has a Sharpe ratio of 1.224, and records the smallest emerging maximum drawdown at -11.66%. These are factor-level gross results, so the large emerging improvement should still be viewed alongside implementation frictions not represented in the source data.

Table 6. Strategy performance with strictly ex-ante scaling

Panel A. Full-sample performance

Market	Strategy	N	Mean ann.	Vol ann.	Sharpe	Max DD
Developed	Static	426	6.85%	12.85%	0.533	-43.61%
Developed	Vol-managed	414	6.52%	7.54%	0.865	-10.74%
Developed	Dynamic	342	10.03%	12.21%	0.821	-26.18%
Emerging	Static	436	10.24%	10.45%	0.980	-37.26%
Emerging	Vol-managed	424	15.65%	12.14%	1.289	-15.45%
Emerging	Dynamic	352	11.31%	9.24%	1.224	-11.66%

Panel B. Full-sample tail statistics

Market	Strategy	Skewness	VaR 5%	CVaR 5%
Developed	Static	-0.932	-5.01%	-9.32%
Developed	Vol-managed	0.934	-2.28%	-4.15%
Developed	Dynamic	0.878	-4.72%	-6.66%
Emerging	Static	-0.991	-3.67%	-7.24%
Emerging	Vol-managed	1.017	-3.88%	-5.97%
Emerging	Dynamic	0.275	-2.72%	-5.22%

Panel C. Subperiod performance

Market	Period	Strategy	N	Mean ann.	Vol ann.	Sharpe	Max DD
Developed	Pre-2008	Static	206	9.87%	13.69%	0.721	-43.05%
Developed	Pre-2008	Vol-managed	194	8.74%	7.79%	1.122	-10.74%
Developed	Pre-2008	Dynamic	122	14.29%	14.58%	0.980	-26.18%
Developed	Post-2010	Static	196	6.45%	9.57%	0.674	-20.10%
Developed	Post-2010	Vol-managed	196	5.31%	7.42%	0.715	-8.82%
Developed	Post-2010	Dynamic	196	6.36%	9.07%	0.701	-11.91%
Emerging	Pre-2008	Static	216	10.94%	10.35%	1.056	-21.61%
Emerging	Pre-2008	Vol-managed	204	20.05%	13.81%	1.452	-14.59%
Emerging	Pre-2008	Dynamic	132	12.87%	10.30%	1.249	-11.66%
Emerging	Post-2010	Static	196	12.48%	9.20%	1.356	-14.52%
Emerging	Post-2010	Vol-managed	196	13.34%	10.34%	1.290	-12.91%
Emerging	Post-2010	Dynamic	196	10.72%	8.89%	1.205	-11.23%

Notes: Vol-managed uses expanding inverse-variance scaling; Dynamic uses the expanding state forecast. Both cap leverage at 3.

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Figures 4 and 5 show cumulative strategy wealth. The visual message is not that any one managed rule dominates at every date. Rather, scaling changes the path of returns. In developed markets, volatility management removes much of the severe drawdown behaviour that makes static momentum difficult to hold. In emerging markets, both managed rules smooth the path, but their relative advantage changes across time.

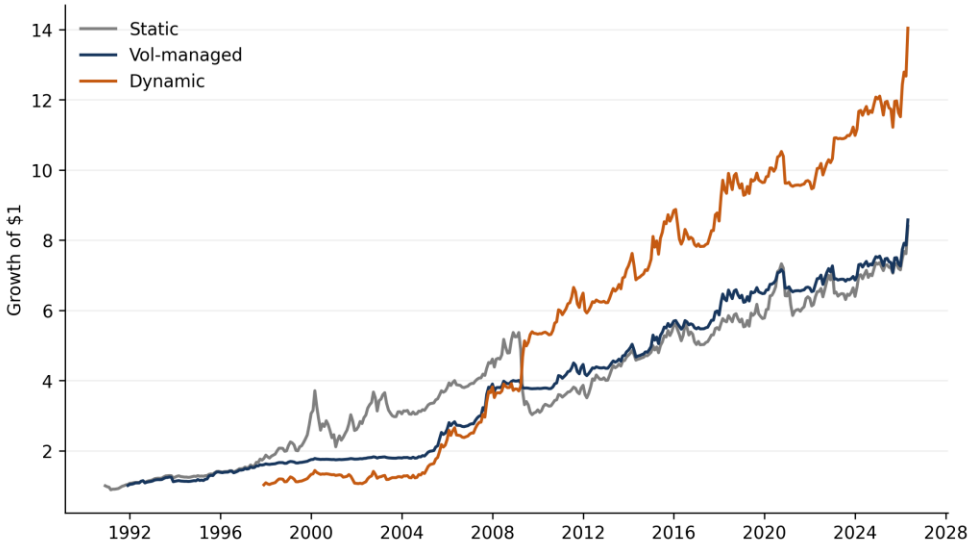


Figure 4. Developed-market cumulative wealth: static, volatility-managed and dynamic momentum.

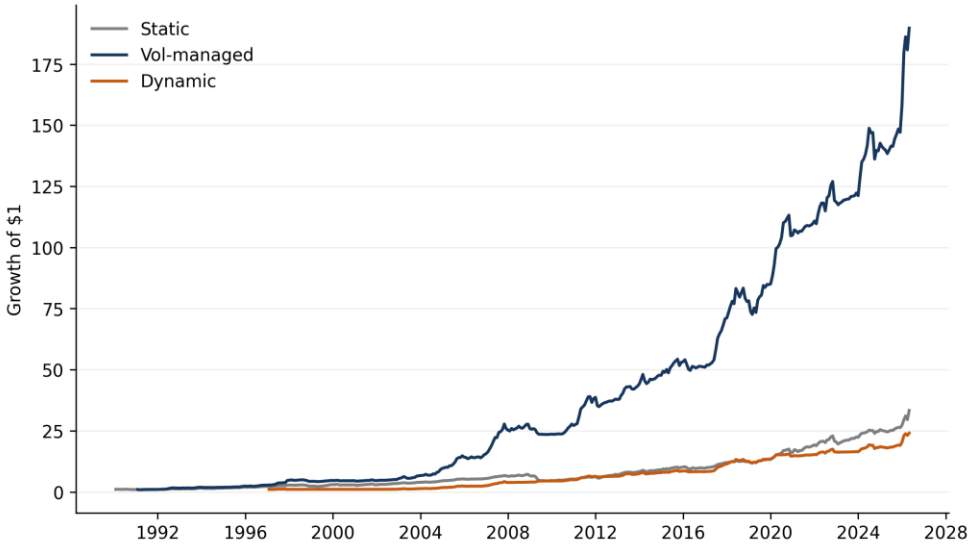


Figure 5. Emerging-market cumulative wealth: static, volatility-managed and dynamic momentum.

Panel C of Table 6 makes the instability explicit. Before 2008, developed volatility management raises the Sharpe ratio from 0.721 to 1.122 and reduces maximum drawdown from -43.05% to -10.74%. Emerging volatility management raises the pre-2008 Sharpe from 1.056 to 1.452, although its higher exposure also produces higher annualised volatility than static WML. The pre-2008 period is therefore the setting in which the managed rules provide the clearest combination of better risk-adjusted performance and smaller drawdown exposure.

After 2010, the pattern is more mixed. Developed volatility management still halves maximum drawdown from -20.10% to -8.82% and modestly raises the Sharpe ratio from 0.674 to 0.715, despite lowering mean return. The emerging result is weaker: static WML has a Sharpe ratio of 1.356, compared with 1.290 for volatility management and 1.205 for the dynamic rule. Drawdowns still improve, but only modestly for the volatility-managed strategy. This is direct evidence against a time-invariant claim that volatility management always improves risk-adjusted performance.

4.6 Robustness and post-2010 stability

Table 7 collects the robustness evidence in three panels. Panel A re-estimates Equation (6) by subperiod. Developed markets before 2008 show a large negative bear × volatility coefficient of -3.1155 with a test statistic of -2.74. After 2010 the same coefficient is only -0.2312 and statistically insignificant. Emerging interactions are not significant in either subperiod and become positive after 2010. The attenuation is therefore not a minor numerical detail; the aggregate state relation is concentrated in the earlier developed-market sample.

Panel B changes the market-volatility window. Developed interaction coefficients remain negative for 6-, 12- and 24-month volatility, but statistical strength is greatest at 12 months and disappears at 24 months. Emerging interactions are also negative across the three windows, with the 6-month estimate marginally significant but the 12- and 24-month estimates imprecise. The conclusion is therefore about a family of state-dependent patterns, not a parameter invariant to the measurement horizon.

Panel C returns to the direct crash model. At the unconditional 10% tail, developed and emerging volatility coefficients are positive, but the bear × volatility interactions are not significant. More importantly, the rolling 60-month 5% crash definition changes the baseline developed coefficient pattern: volatility remains positive and significant, the interaction becomes positive and marginally

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significant, and the bear indicator is negative. Emerging rolling-threshold coefficients remain imprecise. These results demonstrate why the paper does not elevate the baseline logit into a universal ex-ante crash predictor. The qualitative message that tail risk varies with market conditions survives, but the exact coefficient decomposition depends on how a crash is defined.

Table 7. Robustness and stability checks

Panel A. Subperiod conditional-return regressions

Market	Period	Bear	Vol	Bear × Vol	N	Adj. R^2
Developed	Pre-2008	0.1593*** (2.72)	-0.0589 (-0.24)	-3.1155*** (-2.74)	182	0.004
Developed	Post-2010	0.0030 (0.08)	-0.1258 (-0.78)	-0.2312 (-0.30)	196	0.005
Emerging	Pre-2008	0.0141 (0.73)	-0.1852 (-1.11)	-0.1913 (-0.57)	192	0.009
Emerging	Post-2010	-0.0261 (-1.56)	-0.0105 (-0.08)	0.3341 (1.03)	196	0.016

Panel B. Alternative market-volatility windows

Market	Volatility window	Bear × Vol	N
Developed	6 months	-0.8660* (-1.65)	402
Emerging	6 months	-0.3478* (-1.83)	412
Developed	12 months	-0.9672** (-2.02)	402
Emerging	12 months	-0.2594 (-1.26)	412
Developed	24 months	-0.3281 (-0.53)	402
Emerging	24 months	-0.2575 (-1.00)	412

Panel C. Alternative direct crash-probability models

Market	Crash definition	Bear	Vol	Bear × Vol	N	Crashes
Developed	10% unconditional tail	1.2556 (1.12)	38.6271*** (3.14)	-17.8313 (-0.91)	402	41
Developed	5% rolling 60-month tail	-3.7695** (-2.47)	41.2515*** (3.22)	38.0917* (1.77)	366	30
Emerging	10% unconditional tail	0.4665 (0.43)	21.9215* (1.80)	-1.7056 (-0.10)	412	39
Emerging	5% rolling 60-month tail	-0.2526 (-0.17)	11.2068 (0.54)	13.4058 (0.55)	376	27

*Notes: HAC test statistics in parentheses. ***, ** and * denote 1%, 5% and 10% significance.*

The post-2010 evidence admits several possible economic explanations, but the data do not identify one uniquely. Greater awareness of momentum crashes, broader volatility targeting, changes in market microstructure and the institutionalisation of factor investing could all alter the relation between state variables and subsequent returns. The replication literature also cautions that factor results can be sensitive to sample endpoints and empirical choices (Harvey et al. 2016; Hou et al. 2020; Jensen et al. 2023). The appropriate conclusion is therefore empirical rather than causal: the state signal and the managed-strategy advantage are not stable through time, and emerging-market post-2010 performance is the clearest case in which full-sample improvements should not be extrapolated mechanically.

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This instability is not merely a reason to weaken the conclusion; it changes the economic use of the model. A stable forecasting law could be embedded as a permanent allocation rule. An unstable state relation is better treated as one input into a risk-governance process in which exposure rules are monitored against recent realised behaviour. The post-2010 evidence therefore supports periodic validation rather than abandonment: managers can retain a transparent volatility overlay when drawdown control is valuable, but should not assume that a historical Sharpe improvement is a permanent property of the strategy. That interpretation also avoids using the pre-2008 crisis-prone sample to justify aggressive leverage in a later regime where the conditional relationship is weaker.

5. Discussion and implications

The first implication is methodological. Conditional WML returns and the probability of a lower-tail event are related but distinct outcomes. The continuous-return model therefore characterises variation in the conditional mean, while the binary specification examines crash incidence directly. The resulting interpretation is deliberately limited: lagged market states contain information about subsequent momentum weakness and, in some specifications, crash probability, but the strength and functional form of this relation vary across markets and crash definitions.

Second, the worst return episodes remain economically informative even when forecasting power is qualified. The April-May 2009 observations illustrate why momentum can be especially difficult to manage: large factor losses occurred while aggregate markets rebounded sharply. This pattern is consistent with the short-leg rebound mechanism in Daniel and Moskowitz (2016), whereby past losers recover strongly after prolonged market weakness. Risk monitoring should therefore combine prior market decline, volatility and rebound conditions rather than rely only on realised momentum drawdowns.

Third, volatility management is best viewed as a risk-budgeting overlay rather than a return-forecasting rule (Feng and Yang 2026). In developed markets, the ex-ante strategy does not increase average return, but it materially reduces volatility, tail losses and maximum drawdown. The fall in historical maximum drawdown from 43.61% to 10.74% is economically important for institutions facing leverage constraints, redemption risk or drawdown limits. This interpretation is consistent with the risk-management motivation of Barroso and Santa-Clara (2015) and Moreira and Muir

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(2017), while remaining consistent with the cautions raised by Cederburg et al. (2020) and DeMiguel et al. (2024).

The emerging-market evidence is more mixed. Static WML earns a higher average return and Sharpe ratio than developed WML, but the aggregate crash-probability coefficients are less precise, while the rebound regression is stronger. This suggests that contemporaneous recovery dynamics may be more informative than lagged aggregate state variables in emerging markets. Moreover, the factor returns do not incorporate the larger trading costs, short-sale constraints and market-impact effects that may arise in practice. The 15.65% gross annualised return of the volatility-managed emerging strategy should therefore be interpreted as factor-level evidence rather than as a directly achievable net return.

Post-2010 results further limit any universal claim for volatility scaling. In emerging markets, static momentum has the highest post-2010 Sharpe ratio among the three strategies, although the managed rules still reduce drawdown. The value of the overlay therefore depends on the investor's objective. A drawdown-sensitive institution may still prefer volatility management, whereas an investor focused only on realised Sharpe performance may not. This instability supports regular monitoring and recalibration rather than mechanical reliance on a historically successful timing rule.

The international comparison also requires restraint. The pooled model does not reject the emerging-specific slope adjustments jointly, but this does not establish identical underlying structures. Statistical strength differs across markets: developed evidence is more closely associated with the lagged bear-volatility interaction, while emerging evidence is stronger in rebound conditions. A common risk-monitoring framework may therefore be useful, but region-specific calibration and stress testing remain appropriate. For portfolio construction, the most defensible common workflow is to monitor prior market state and volatility, recognise the risk of rebound-driven momentum reversals, and evaluate the overlay separately across regions and regimes.

Several limitations remain. The analysis uses aggregate market-group factors rather than country- or security-level portfolios, so it cannot identify the constituents driving individual crash episodes. Factor returns exclude transaction costs, financing constraints, taxes and capacity effects. The baseline crash threshold is estimated from the unconditional sample, although the rolling-threshold robustness addresses this concern. The dynamic strategy uses only three state variables; richer predictors might improve forecasting but would also increase data-mining risk. Finally, the analysis

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documents conditional associations and historical portfolio rules rather than a causal structural model of momentum crashes. These limitations narrow the interpretation but do not eliminate the practical value of transparent, state-aware risk management.

6. Conclusion

This paper examines momentum crash risk, market states and volatility management in developed and emerging equity markets through April 2026. The analysis separates three questions that should not be conflated. Continuous WML regressions show how momentum returns vary with lagged states; binary logits directly model lower-tail events; and portfolio tests evaluate whether transparent ex-ante scaling changes realised investment risk.

The strongest aggregate state evidence appears in developed markets and in the earlier sample. Emerging markets show clear crash episodes and strong rebound effects, but weaker precision in the lagged crash-probability models. Cross-market interaction tests fail to reject equality of several slopes, but this is interpreted as insufficient evidence of difference rather than proof of identical structure. Robustness tests also show that crash definitions matter.

Volatility management remains economically useful, especially as a drawdown-control device. With expanding ex-ante normalisation, developed maximum drawdown falls from 43.61% to 10.74% and emerging maximum drawdown falls from 37.26% to 15.45%. Yet the advantage is not stable: after 2010, particularly in emerging markets, static momentum has the higher Sharpe ratio. The central practical conclusion is therefore conditional rather than universal. Momentum can remain a valuable return source, but investors should combine it with explicit downside-risk governance and should reassess any timing rule as market structure and realised performance change.

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Data Availability

All data used in this study are publicly available from Kenneth R. French's Data Library.

A **replication package**[2], including the source data and a ready-to-run Python Jupyter Notebook that reproduces the empirical results, tables, and figures, is publicly available at the GitHub repository.

Declaration of generative AI use

Generative artificial-intelligence tool – GPT5.6 sol model - were used for language editing, presentation support, and enhancing & checking analysis codes; all empirical analysis is reproducible from the supplied data and notebook. All mathematical derivations, interpretations,

[2] https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/AE_Momentum_Crashes_across_Developed_and_Emerging_Markets

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numerical specifications and final content were reviewed and verified by the author, who assumes full responsibility for the article.

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[i] https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research

[ii] https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library

[iii] https://www.linkedin.com/in/dr-xuan-feng-217556a4/

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